

Properties of real numbers



1

- | | | | |
|---------|--------|---------|--------|
| a True | b True | c True | d True |
| e False | f True | g True | h True |
| i True | j True | k False | |

2 False

3 False

4

- | | | | |
|--------|--------|--------|--------|
| a True | b True | c True | d True |
|--------|--------|--------|--------|

5

- | | |
|---------------------------------------|-------------------------------|
| a $0 + 9m = 9m$ | b $1 \times 4b = 4b$ |
| c $3d - 0 = 3d$ | d $a \times 19 = 19 \times a$ |
| e $d + \frac{1}{7} = \frac{1}{7} + d$ | |

6

- | | |
|-------------------------|-------------------------|
| a $5(s + 6) = 5(6 + s)$ | b $8(4 + m) = 8(m + 4)$ |
|-------------------------|-------------------------|

7

- | | | | |
|------|------|-------|------|
| a No | b No | c Yes | d No |
|------|------|-------|------|

8 $\frac{1}{2x}$

9

- | | |
|------------------------|------------------------|
| a Commutative property | b Commutative property |
| c Commutative property | d Associative property |
| e Associative property | f Associative property |
| g Identity property | h Inverse property |
| i Inverse property | j Inverse property |
| k Identity property | l Identity property |
| m Commutative property | n Commutative property |
| o Commutative property | p Associative property |



10

- a Commutative property
- c Inverse property
- e Associative property
- g Commutative property
- i Identity property

- b Associative property
- d Associative property
- f Associative property
- h Associative property

11 Multiplication property of equality

12

- a Yes
- b Yes
- c Yes

13

- a
 - i Commutative property of addition
 - ii $-7 + 8 = 8 + (-7)$
- b
 - i Commutative property of multiplication
 - ii $-6 \times 5 = 5 \times (-6)$
- c
 - i Associative property of addition
 - ii $(9 + 7) + 6 = 9 + (7 + 6)$
- d
 - i Associative property of multiplication
 - ii $6 \times (9 \times 7) = (6 \times 9) \times 7$

14

- a $3 + x$
- b $q \times 4$
- c $d \times g$

15

- a $2 \times (a \times b) = (2 \times a) \times b$
- b $(ab)c = a(bc)$
- c $9 + (a + b) = (9 + a) + b$

16 $8(5h) = (8 \times 5)h$
 $= 40h$

17 $3 \times 6 + 3 \times 5$

18 The sum of a real number and its additive inverse is equal to 0, the additive identity.

19 False, because the commutative property allows us to swap added terms but not the coefficients of those terms.



20

- a Distributive property
- b Commutative property of addition
- c Associative property of addition
- d Commutative property of addition

Additional questions

21

- a Commutative property, reordering
- b Commutative property, reordering
- c Reordering
- d Associative property, commutative property, reordering
- e Distributive property
- f Associative property, commutative property, reordering

22

- a True
- b False
- c False
- d True

23

- a 75
- b 58

24

- a $27 + 13$
- b 88

25 260

26 79

27

- a Yes
- b Reordering

28 2

29

- a 26
- b 12

30 11



31

a Yes

b Reordering

32

a 4×5

b $4 \times 13 \times 5 = 4 \times 5 \times 13$ or $4 \times 13 \times 5 = 5 \times 4 \times 13$

33 340

34

a 3

b 20

c Distributive property

d 805

35

a 30

b 1

c Distributive property

d 754

36

a 260

b 26

c 286

37

a 320

b 32

c 288

38

a 8982

b 7014

c 1364

d 2472

39 4

40

a 28

b 28

c Distributive property

d 24

41

a 21

b 7

c Distributive property

d 17

42

a 202

b 99



c 87

d 45